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Next item →

Simulation Techniques for Bayesian Inference Quiz

Instructions

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

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Assignment details

Submitted	**Attempts**
Dec 9, 1:58 AM IST	Unlimited

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1. Why might we use importance resampling? **1 / 1 point**

- (●) To convert weighted samples into unweighted samples
- ( ) To avoid the use of MCMC methods
- ( ) To simplify the posterior distribution
- ( ) To generate dependent samples

Correct
Correct! Importance resampling converts weighted samples into unweighted samples, facilitating further analysis.

2. What is a benefit of sampling without replacement in importance resampling? **1 / 1 point**

- ( ) It increases computational complexity
- (●) It provides a more desirable approximation between the starting and target densities
- ( ) It avoids the need for prior distributions
- ( ) It ensures all weights are equal

Correct
Correct! Sampling without replacement yields a more desirable intermediate approximation between the starting and target densities.

3. What is rejection sampling? **1 / 1 point**

- (●) A method to draw samples from an easy-to-sample distribution and reject some to match the target distribution
- ( ) A method to ensure all samples are accepted
- ( ) A method that avoids using a target distribution
- ( ) A method that always improves computational efficiency

Correct
Correct! Rejection sampling involves drawing samples from a proposal distribution and accepting or rejecting them to match the target distribution.